Problem Set 5C

Vector product

Do these after the slides for Section 5C. Textbook Exercise 5C. Answers are on the last page. Show full working in your exercise book.

Skills

1 Find:

(a) $(\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}) \times (4\mathbf{i} + 5\mathbf{j} + 6\mathbf{k})$

(b) $(2\mathbf{i} - \mathbf{k}) \times (\mathbf{i} + 3\mathbf{j} + 2\mathbf{k})$

(c) $\mathbf{i} \times (\mathbf{j} + \mathbf{k})$

2 Find a unit vector perpendicular to both $\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$ and $2\mathbf{i} + \mathbf{j} - 2\mathbf{k}$.

3 Find the area of the parallelogram with adjacent sides $3\mathbf{i} + \mathbf{j}$ and $\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$.

4 Find the exact area of the triangle with vertices $A(1, 1, 1)$, $B(2, 3, 4)$ and $C(4, 2, 2)$.

5 Simplify:

(a) $\mathbf{a} \times (\mathbf{a} + \mathbf{b})$

(b) $(\mathbf{a} - 2\mathbf{b}) \times (2\mathbf{a} + \mathbf{b})$

(c) $\mathbf{a} \cdot (\mathbf{a} \times \mathbf{b})$

Exam style

6 For the points $A(2, 1, 0)$, $B(3, 3, 2)$ and $C(5, 2, 1)$, find the area of triangle ABC and hence the distance from C to the line AB.

7 The parallelogram with adjacent sides $\mathbf{i} + 2\mathbf{k}$ and $\mathbf{j} + t\mathbf{k}$ has area 3. Find the possible values of t.

8 Given $|\mathbf{a}| = 2$, $|\mathbf{b}| = 5$ and $|\mathbf{a} \times \mathbf{b}| = 8$, find the possible values of $\mathbf{a} \cdot \mathbf{b}$.

Problem Set 5C: answers

1 (a) $-3\mathbf{i} + 6\mathbf{j} - 3\mathbf{k}$ (b) $3\mathbf{i} - 5\mathbf{j} + 6\mathbf{k}$ (c) $\mathbf{k} - \mathbf{j}$

2 $\frac{1}{3}(-2\mathbf{i} + 2\mathbf{j} - \mathbf{k})$ (or its negative)

3 $|2\mathbf{i} - 6\mathbf{j} + 5\mathbf{k}| = \sqrt{65}$

4 $\frac{1}{2}|-\mathbf{i} + 8\mathbf{j} - 5\mathbf{k}| = \dfrac{3\sqrt{10}}{2}$

5 (a) $\mathbf{a} \times \mathbf{b}$ (b) $5(\mathbf{a} \times \mathbf{b})$ (c) 0

6 $\overrightarrow{AB} \times \overrightarrow{AC} = 5\mathbf{j} - 5\mathbf{k}$; area $\dfrac{5\sqrt{2}}{2}$; distance $\dfrac{5\sqrt{2}}{3}$

7 $|-2\mathbf{i} - t\mathbf{j} + \mathbf{k}| = \sqrt{5 + t^2} = 3$, so $t = \pm 2$

8 $\sin\theta = \frac{4}{5}$, $\cos\theta = \pm\frac{3}{5}$, so $\mathbf{a} \cdot \mathbf{b} = \pm 6$